TEXORA · LECTURE SHEET

Electro-spinning and Gel-spinning

Two special spinning routes solve opposite problems: electrospinning makes the finest fibres of all, down to tens of nanometres, while gel spinning makes polyethylene into one of the strongest fibres known. This sheet covers both principles, their parameters and their products.

YM-4203 · TECHNICAL FIBRE & YARN · TOPIC 3 · LEVEL 4 TERM 2 · YARN MANUFACTURING SPECIALIZATION

1. Electrospinning — principle

Electrospinning draws a fibre from a polymer solution (or, less commonly, a melt) using electrostatic force instead of mechanical drawing. A high voltage — typically 10–30 kV — is applied between a spinneret needle and a collector. The droplet of solution at the needle tip becomes charged; repulsion between its surface charges fights its surface tension. At a critical voltage the droplet deforms into a cone, the **Taylor cone**, and a fine charged jet issues from its tip.

The jet first travels straight for a short distance, then becomes unstable and bends into a rapidly whipping, spiralling path. During this whipping the jet is stretched enormously and the solvent evaporates, so it lands on the collector as a solid fibre with a diameter typically between about 50 nm and 1 µm — one to two orders of magnitude finer than a conventional filament. The product is normally collected as a random nonwoven web.

1.1 The setup, step by step

1. A **syringe pump** feeds polymer solution at a controlled, very low flow rate through tubing to a metal **spinneret needle**.
2. A **high-voltage supply** connects its positive terminal (the usual laboratory arrangement) to the needle and its negative terminal / earth to the collector; polarity can be reversed in some work, but one electrode is charged and the collector is grounded.
3. The charged droplet forms a **Taylor cone**; a jet ejects, whips, dries and thins in flight.
4. Fibres deposit on the **grounded collector** — a flat plate for random webs, or a rotating drum or disc when partial alignment is wanted.
5. The web is removed, dried fully to clear residual solvent, and tested (fibre diameter distribution, web uniformity, function).

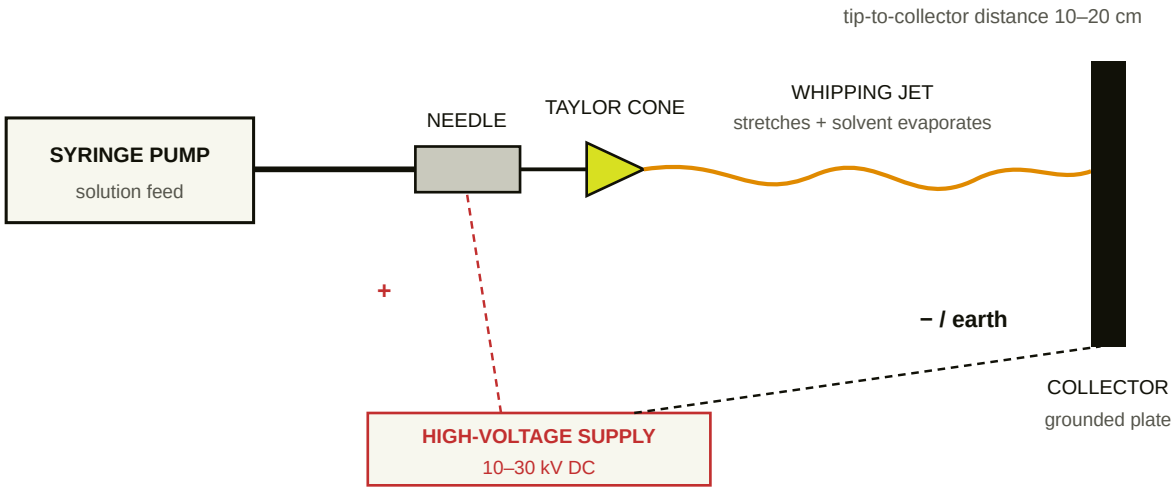


Figure 1 — Electrospinning setup. The needle is held at high positive potential, the collector is grounded; a Taylor cone forms at the needle tip and the whipping jet dries into nanofibres.

2. Electrospinning parameters and their effects

PARAMETER	INCREASE CAUSES	NOTES / LIMITS
Voltage	Stronger jet stretching; often finer fibres up to a point	Too high: multiple jets, beads, unstable cone
Flow rate	More material per unit time; coarser fibres	Too high: wet fibres, beads, dripping — solvent cannot evaporate in flight
Tip-to-collector distance	More drying and whipping time; finer, drier fibres	Too far: field weakens (E = V/d), deposition spreads and is lost
Solution concentration / viscosity	Thicker, continuous fibres; beads disappear above a threshold	Too low: electrospraying / beads-on-string; too high: needle blocks
Solvent volatility	Faster drying in flight	Too volatile: skin forms at the needle and blocks it
Humidity / temperature	Change evaporation and surface texture (pores)	Must be recorded for reproducible work

The governing idea is a balance: enough charge to draw the jet, enough polymer entanglement to keep it continuous, and enough flight time to dry it. **Beads** are the signature fault of an unbalanced process — usually low concentration, excessive flow or unsuitable voltage — and a beaded web is not a nanofibre product, whatever the label says.

2.1 Applications

- **Filtration** — a thin nanofibre layer on a conventional substrate captures fine particles at low pressure drop.
- **Wound dressings** — fine, porous, high-surface webs that can carry antimicrobial agents and conform to the wound.
- **Tissue scaffolds** — webs whose fibre scale resembles the extracellular matrix, supporting cell attachment in research and regenerative medicine.
- **Protective clothing and sensors** — breathable barrier layers and functionalised fibre mats.

The honest limitation is throughput: a single needle produces grams per hour, not kilograms. Multi-needle and needleless systems raise output and are an active engineering field, but electrospinning remains a speciality, high-value process.

3. Worked example — the electric field

> **WORKED EXAMPLE — FIELD STRENGTH E = V / D**
>
> Applied voltage V = 20 kV = 20,000 V; tip-to-collector distance d = 15 cm = 0.15 m.
> E = V / d = 20,000 / 0.15 = 133,333 V/m ≈ **133 kV/m**.
> This is why distance matters as much as the voltage dial: doubling d at the same voltage halves the field that forms the cone and draws the jet, so voltage and distance must always be reported together.

4. Gel spinning — the route to ultra-strong polyethylene

Gel spinning was developed to exploit UHMWPE, whose molecules are so long (molecular weight in the millions) that a normal melt is far too viscous to extrude and a normal solution is a tangle that cannot be drawn. The route is:

1. **Dissolve** a small percentage of UHMWPE in a solvent (historically decalin or paraffin oil) at elevated temperature, making a homogeneous solution.
2. **Extrude** the solution through a spinneret; on cooling it sets to a **gel filament** in which the chains are disentangled compared with the melt, held apart by solvent.
3. **Extract** the solvent (or evaporate it, depending on the solvent system), leaving a porous filament of disentangled, very long chains.
4. **Ultra-draw** the filament at a temperature just below its melting point, at draw ratios far beyond normal practice — tens of times — so the long chains unfold, align almost perfectly along the fibre axis and crystallise in extended-chain form.

The logic is the central lesson of fibre science: strength comes from long, aligned, load-sharing molecules. Ordinary polyethylene is weak because its chains are folded and tangled, so load is carried by weak inter-chain forces. Gel spinning removes the tangles *before* drawing, so the draw can align the chains themselves, and the final fibre's strength and modulus approach what the covalent chain can deliver: tenacity typically 3–4 GPa at a density of only 0.97 g/cm³ (Topic 1). The price of this perfection is a low melting point (about 145–150 °C) and creep under sustained load, inherited from polyethylene chemistry.

WORKED EXAMPLE — WHAT A DRAW RATIO DOES TO LINEAR DENSITY

Gel spinning draw ratio = 40. An extracted gel filament of 400 dtex is drawn uniformly.

Drawing stretches length by 40 while mass is unchanged, so linear density falls by 40:

Final linear density = 400 / 40 = **10 dtex**.

The same conservation idea applies in every drawing process: linear density after drawing = linear density before / draw ratio (neglecting any material loss).

5. Comparison of the three spinning routes

FEATURE	MELT SPINNING	ELECTROSPINNING	GEL SPINNING
Feed	Polymer melt	Polymer solution	Dilute UHMWPE solution → gel
Drawing force	Mechanical take-up + draw rolls	Electrostatic (10–30 kV)	Mechanical ultra-drawing
Typical fibre size	10–30 µm filaments	~50 nm – 1 µm	~10–20 µm filaments
Product form	Continuous filament / staple	Random nonwoven web	Continuous high-strength filament
Signature strength	Speed, scale, low cost	Finest fibres, huge surface area	Highest specific tenacity
Signature weakness	Needs a meltable, stable polymer	Very low throughput; solvent handling	Solvent recovery; product melts ~150 °C
Typical products	Apparel polyester and nylon	Filters, dressings, scaffolds	Ropes, ballistics, cut protection

Glossary

Taylor cone — conical shape a charged droplet takes before jet ejection. **Whipping jet** — bending, spiralling motion that stretches and dries the electrospinning jet. **Beads** — droplet defects in an electrospun web from an unbalanced process. **Gel filament** — solvent-swollen filament set from a polymer solution. **Draw ratio** — final length divided by initial length in drawing. **Extended-chain crystal** — crystal form with chains aligned straight along the fibre axis. **UHMWPE** — ultra-high-molecular-weight polyethylene.

Key points

• Electrospinning: high voltage forms a Taylor cone; the whipping jet dries into fibres of roughly 50 nm – 1 µm, collected as a web.
• Report voltage and distance together — the field E = V/d is what acts on the jet.
• Beads mean imbalance: check concentration, flow rate and voltage first.
• Gel spinning disentangles UHMWPE chains in solution, then ultra-draws them into near-perfect alignment — hence exceptional specific strength.
• Drawing divides linear density by the draw ratio.
• Neither process competes with melt spinning on cost or scale; both win on a property melt spinning cannot reach.

Practice

1. Calculate the field strength when 15 kV is applied over a tip-to-collector distance of 12 cm. Compare it with the worked example and state which setup has the stronger field.
2. An electrospun web is heavily beaded. List three parameter changes you would try, in order, and the reason for each.
3. A gel filament of 600 dtex is drawn at a draw ratio of 30. Calculate its final linear density.
4. Explain why UHMWPE cannot simply be melt spun and conventionally drawn to the same strength as the gel-spun fibre.

Answer key

1. E = 15,000 / 0.12 = **125,000 V/m = 125 kV/m**. This is **lower** than the worked example's 133 kV/m, despite the similar voltage, because the comparison must use V/d, not V alone — here V fell proportionally more than d.
2. (1) **Raise solution concentration/viscosity** — more chain entanglement holds the jet together instead of breaking into droplets. (2) **Reduce flow rate** — less material than the field can draw is a common cause of dripping and beads. (3) **Adjust voltage** (usually raise moderately) — strengthen jet stretching; if beads persist, check distance and solvent choice.

3. Final linear density = 600 / 30 = **20 dtex**.
4. UHMWPE chains are so long that the melt is extremely viscous and heavily entangled; it cannot be extruded and drawn normally, and drawing an entangled structure aligns only segments between tangles, so load is still carried largely by weak inter-chain forces and the fibre breaks before high alignment is reached. Dissolving the polymer disentangles the chains first, so gel spinning's ultra-drawing aligns the chains themselves into extended-chain crystals.
